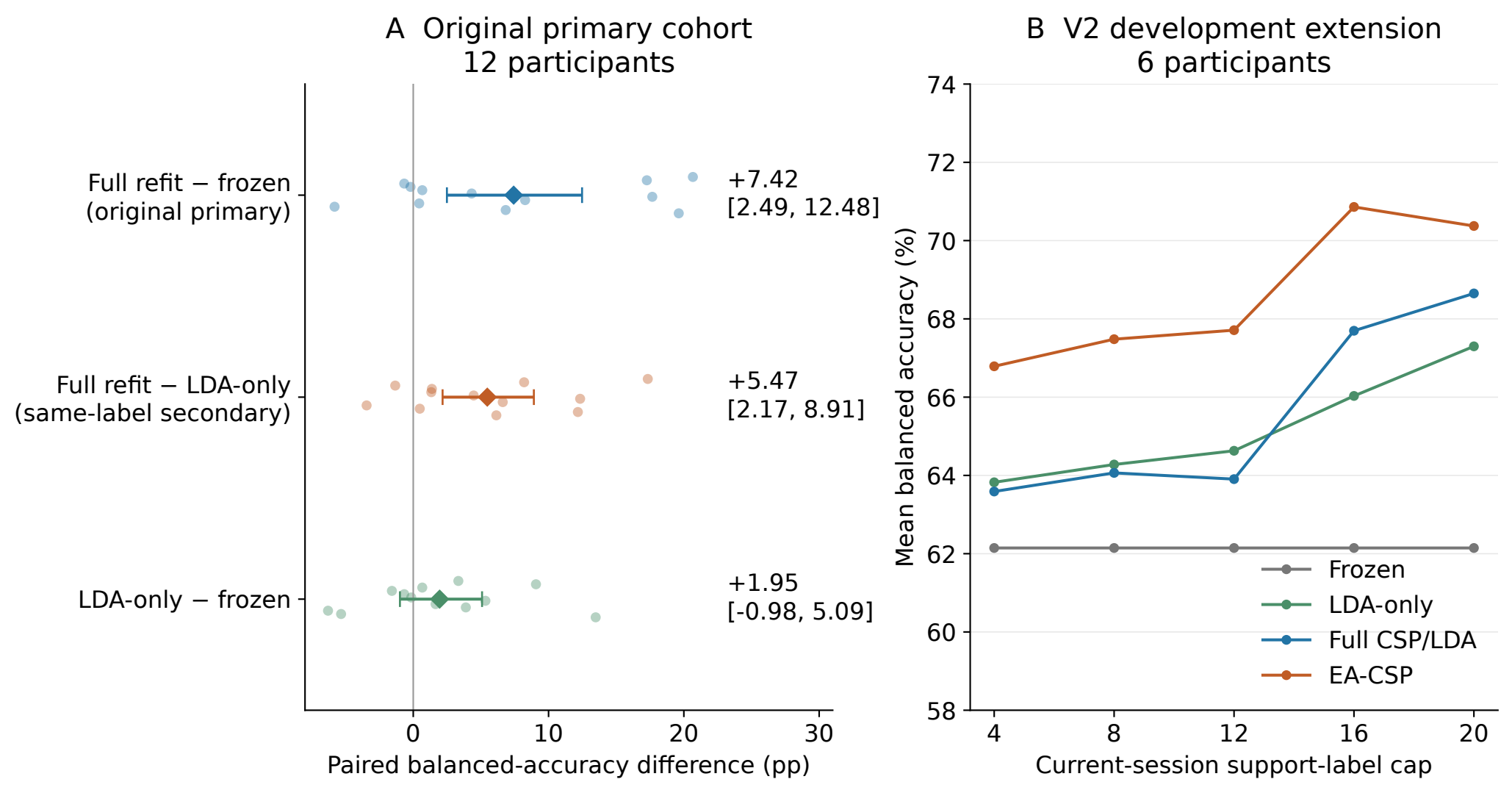
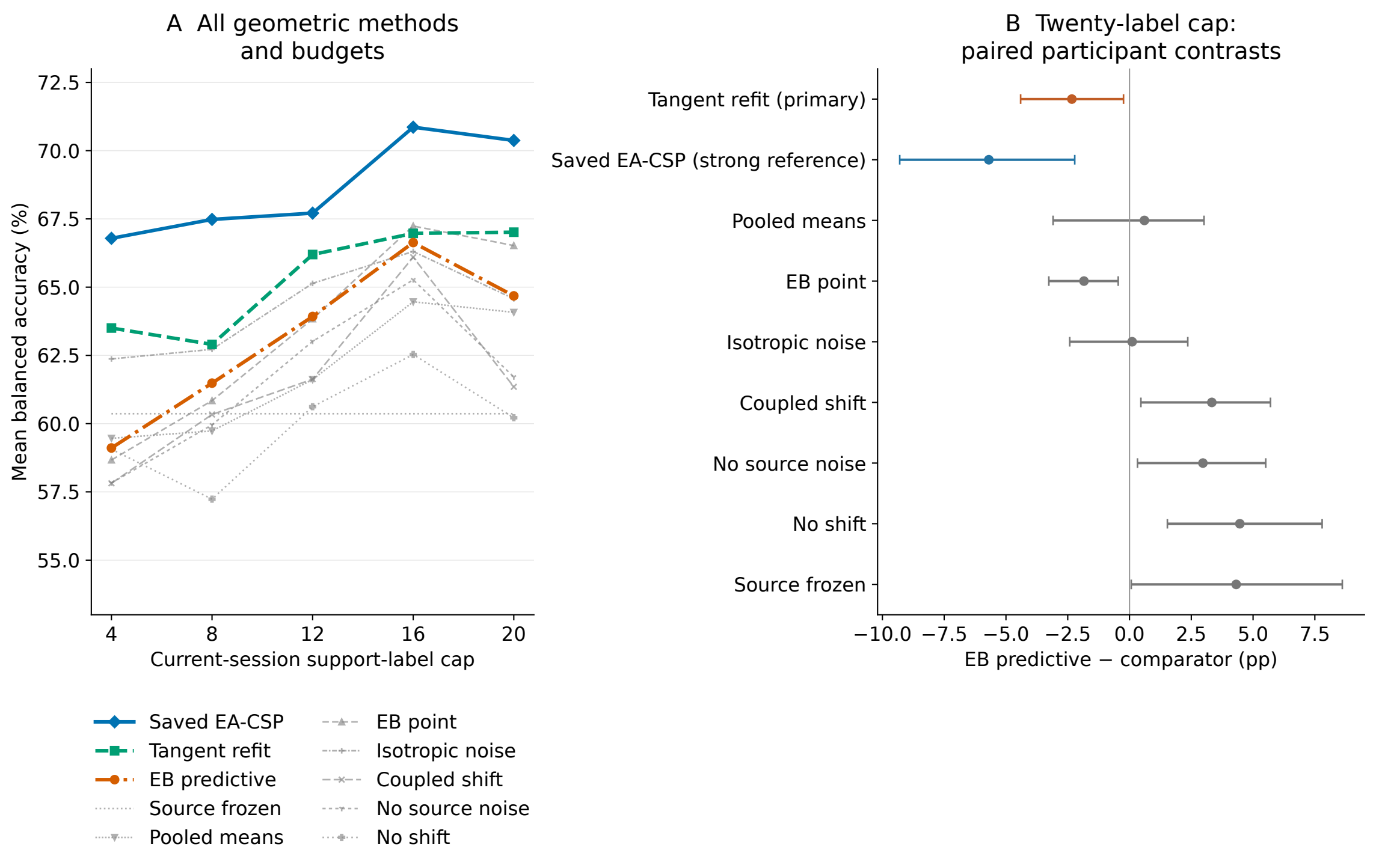
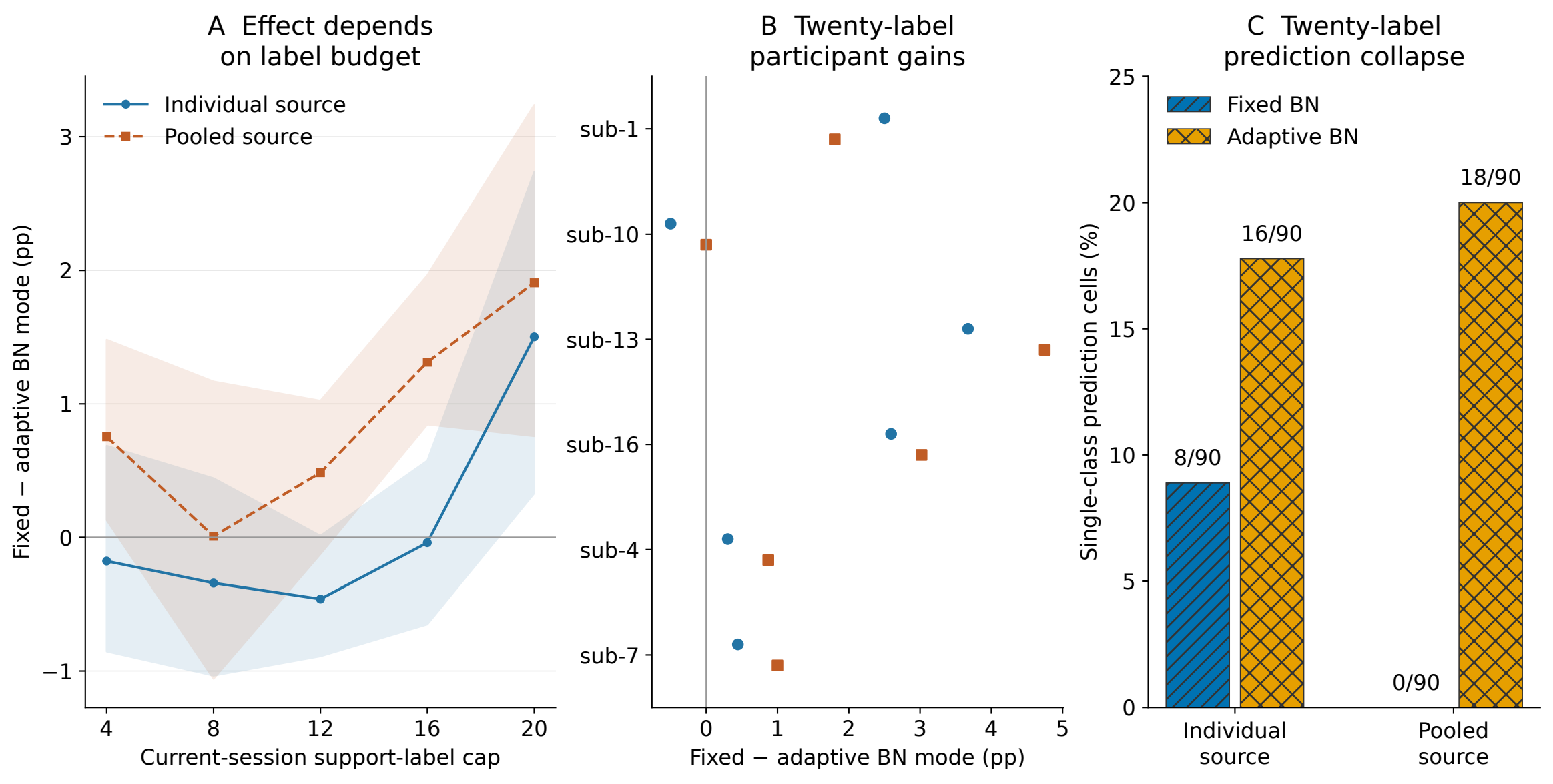
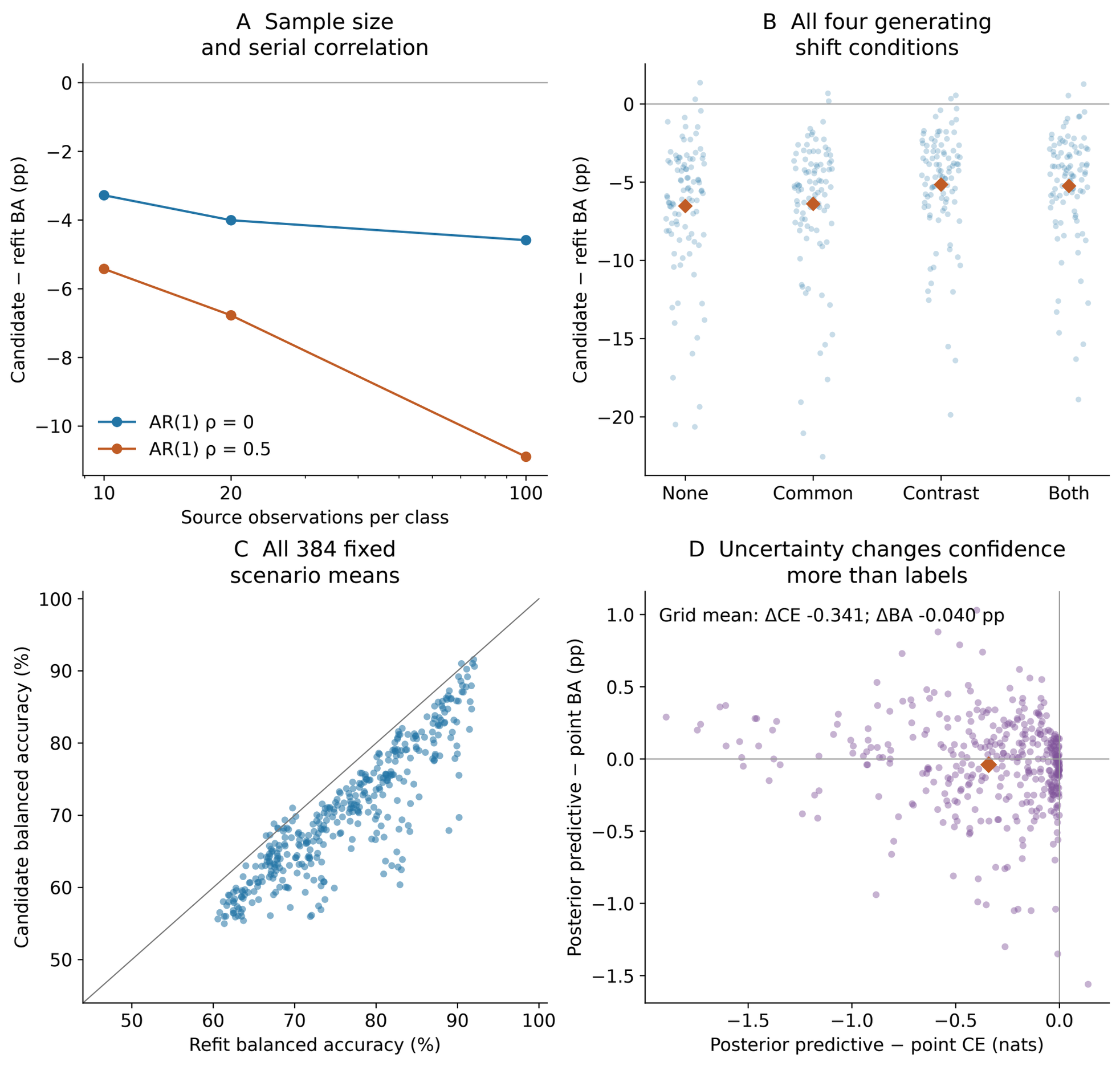
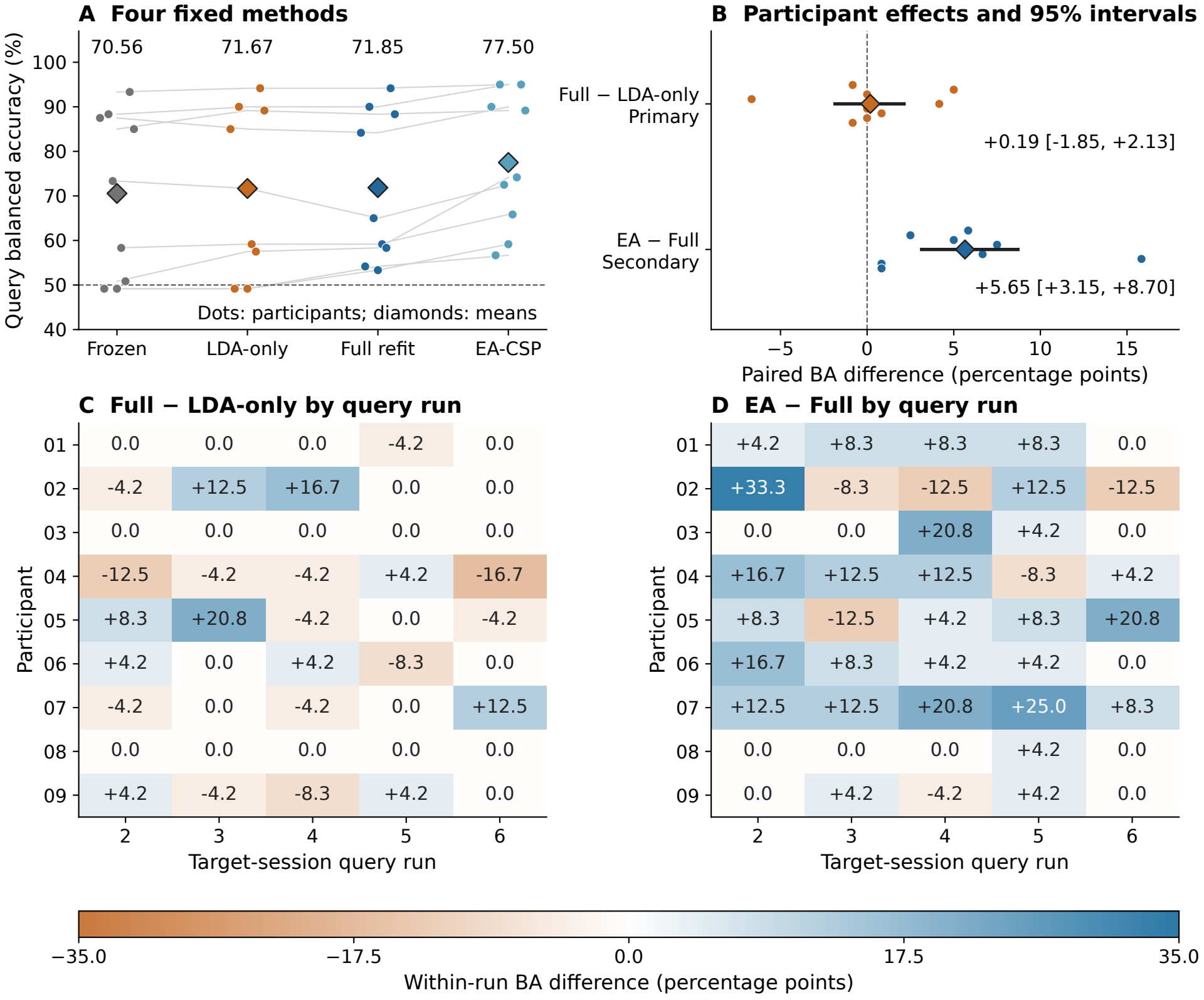

A Original primary cohort
12 participants
Full refit − frozen
(original primary)
+7.42
[2.49, 12.48]
Full refit − LDA-only
(same-label secondary)
+5.47
[2.17, 8.91]
LDA-only − frozen
+1.95
[-0.98, 5.09]
0
10
20
30
Paired balanced-accuracy difference (pp)
B V2 development extension
6 participants
Mean balanced accuracy (%)
74
72
70
68
66
64
62
60
58
4
8
12
16
20
Current-session support-label cap
Frozen
LDA-only
Full CSP/LDA
EA-CSP


A All geometric methods and budgets
Mean balanced accuracy (%)
72.5
70.0
67.5
65.0
62.5
60.0
57.5
55.0
4
8
12
16
20
Current-session support-label cap
Saved EA-CSP
Tangent refit
EB predictive
Source frozen
Pooled means
EB point
Isotropic noise
Coupled shift
No source noise
No shift
B Twenty-label cap: paired participant contrasts
Tangent refit (primary)
Saved EA-CSP (strong reference)
Pooled means
EB point
Isotropic noise
Coupled shift
No source noise
No shift
Source frozen
−10.0
−7.5
−5.0
−2.5
0.0
2.5
5.0
7.5
EB predictive − comparator (pp)


A Effect depends on label budget
Individual source
Pooled source
Fixed − adaptive BN mode (pp)
Current-session support-label cap
B Twenty-label participant gains
sub-1
sub-10
sub-13
sub-16
sub-4
sub-7
Fixed − adaptive BN mode (pp)
C Twenty-label prediction collapse
Fixed BN
Adaptive BN
Single-class prediction cells (%)
8/90
16/90
0/90
18/90
Individual source
Pooled source


A Sample size and serial correlation
Candidate − refit BA (pp)
0
−2
−4
−6
−8
−10
AR(1) ρ = 0
AR(1) ρ = 0.5
10
20
100
Source observations per class
B All four generating shift conditions
Candidate − refit BA (pp)
0
−5
−10
−15
−20
None
Common
Contrast
Both
C All 384 fixed scenario means
Candidate balanced accuracy (%)
100
90
80
70
60
50
50
60
70
80
90
100
Refit balanced accuracy (%)
D Uncertainty changes confidence more than labels
Grid mean: ΔCE -0.341; ΔBA -0.040 pp
Posterior predictive − point BA (pp)
1.0
0.5
0.0
−0.5
−1.0
−1.5
−1.5
−1.0
−0.5
0.0
Posterior predictive − point CE (nats)


A Four fixed methods
70.56
71.67
71.85
77.50
Query balanced accuracy (%)
Dots: participants; diamonds: means
Frozen
LDA-only
Full refit
EA-CSP
B Participant effects and 95% intervals
Full − LDA-only
Primary
+0.19 [-1.85, +2.13]
EA − Full
Secondary
+5.65 [+3.15, +8.70]
Paired BA difference (percentage points)
C Full − LDA-only by query run
Participant
Target-session query run
D EA − Full by query run
Target-session query run
Within-run BA difference (percentage points)